

5 Each meristem cell in a leaf bud is able to grow and divide by mitosis to produce two daughter cells that are genetically identical to each other and to the original dividing cell.

(a) Fig. 5.1 lists the stages in the mitotic cell cycle of a meristem cell.

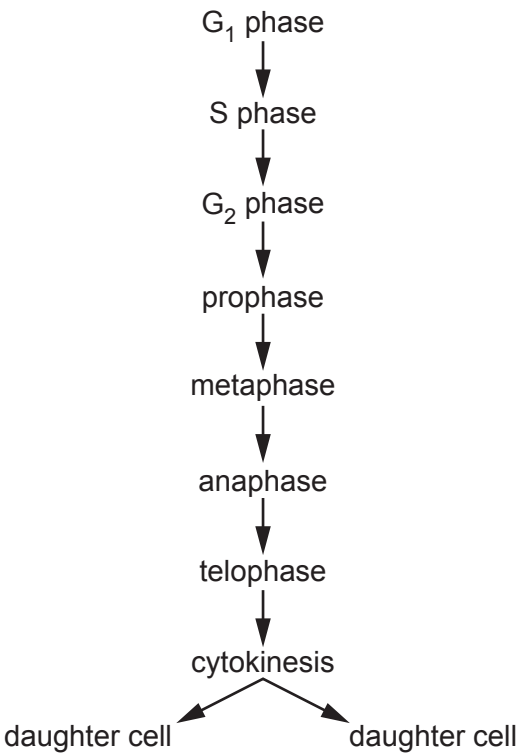


Fig. 5.1

(i) Outline **and** explain the events occurring during S phase, metaphase and anaphase of the mitotic cell cycle that are important in the production of genetically identical daughter cells.

S phase

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metaphase

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anaphase

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- (ii) Immediately after cytokinesis, daughter cells are **not** identical even though they are genetically identical.

Suggest a reason why daughter cells are **not** identical immediately after cytokinesis.

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..... [1]

- (b) Some of the cells resulting from mitotic division in the young leaf form elongated cells that develop into xylem vessel elements. The xylem vessel elements are joined end to end to form xylem vessels.

- (i) Suggest the structural changes that occur when elongated cells develop into xylem vessel elements **and** explain how these changes help xylem vessels to perform their function in transport.

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- (ii) Fig. 5.2 is a plan diagram of a transverse section through a dicotyledonous leaf.

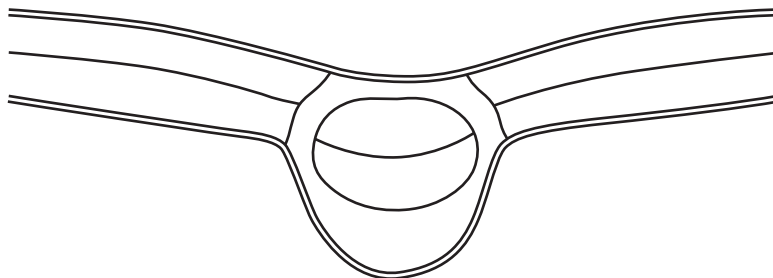


Fig. 5.2

Add a label line and the letter **X** to Fig. 5.2 to identify the location of xylem tissue. [1]

[Total: 10]